

AI Engineering Environment Setup

Session Agenda

Duration: 3 Hours

Session Overview

This session prepares learners entering Machine Learning (ML), Deep Learning (DL), and Generative AI programs to configure and manage the technical environments required for modern AI development. Learners will begin with foundational environment setup concepts, install and configure Python and virtual environments, explore IDEs and Jupyter Notebooks, understand GPU acceleration and cloud-based development options, install essential ML, Deep Learning, and Generative AI libraries, and conclude by building and running their first AI application. The session includes two hands-on labs and a live demonstration to provide practical experience with the complete development workflow.

Learning Outcomes

- Understand the overall AI engineering landscape and how GenAI environments differ from traditional software environments
- Feel comfortable installing and managing Python, virtual environments, and dependencies
- Set up and use Jupyter Notebook and VS Code for AI development
- Understand GPU, CUDA, and cloud notebook platforms such as Google Colab
- Install GenAI libraries and configure API keys, endpoints, and environment variables
- Successfully run a simple LLM application end-to-end
- Recognize and troubleshoot common environment issues using best practices

Detailed Session Schedule

Duration	Topic / Content
Part 1 — Environment Foundations (85 min)	
10 min	Welcome, Session Roadmap & Why This Matters Introduction to the session, what will be covered, and how it connects to the rest of the program.
20 min	AI Engineering Environment Fundamentals What is AI engineering setup; why GenAI environments differ from traditional software environments; CPU vs GPU; local vs cloud execution; why dependency management matters; typical failures in GenAI projects.
20 min	Python & Virtual Environment Setup Python installation, pip basics, virtual environments, package isolation, dependency conflicts, and a brief introduction to Docker.
15 min	Hands-on Lab I: Virtual Environment Setup Learners create a virtual environment, activate it, and install a sample package to see isolation in action.
20 min	IDE, Jupyter & VS Code Setup Why an IDE matters, Jupyter notebook workflow, VS Code setup, Python extensions, and notebook execution.
10 min	Break
Part 2 — GPU, Cloud & Running AI Applications (85 min)	
15 min	GPU, CUDA & Runtime Awareness What is GPU acceleration, VRAM vs RAM, CUDA basics, GPU usage in Google Colab, and why LLMs need GPUs.
10 min	Cloud Notebooks & GPU Platforms Why cloud platforms are used, free GPU options, Colab basics, runtime selection, and connecting Google Drive.
15 min	Hands-on Lab II: Colab GPU Runtime & Drive Learners open Google Colab, select a GPU runtime, and mount Google Drive.
10 min	Installing GenAI Libraries Transformer ecosystem, PyTorch basics, installing AI libraries, and version compatibility.

Duration	Topic / Content
10 min	APIs, Keys & Endpoint Setup What API keys are, endpoint basics, hosted LLMs, authentication, and environment variables.
15 min	Demo: Running First LLM Application Loading transformer models, running inference, prompt-based generation, and a basic text generation workflow.
10 min	Troubleshooting, Recap & Next Steps CUDA mismatch, package conflicts, OOM errors, broken kernels, dependency troubleshooting, notebook restart practices; key takeaways and open Q&A.